

Special Relativity: Gauge Invariance and the Vector Potential

Based on lectures by Leonard Susskind

Lecture notes organized by [LazyingArt LLC](#) with [Video2Book](#)

Original lecture by Leonard Susskind. Notes reconstructed from the lecture and organized by [LazyingArt LLC](#) with [Video2Book](#).

Chapter 1

Gauge Invariance and the Necessity of the Vector Potential

Before beginning the physics, there are two practical matters. The next meeting will be skipped while the lecturer is in Belgium, but the course will continue until all ten lectures have been given. The handwritten notes for the evening have also been prepared at short notice; anyone willing to scan them for the class website will rescue them from a remarkably slow scanner.

The previous lecture was mathematically dense. Rather than pretending otherwise, we shall go through its central calculation again. The aim of the course is a genuine theoretical minimum: enough structure to continue to the next level. If the mathematics could be removed without losing the physics, there would be every reason to remove it. But exact laws require quantitative statements, and quantitative statements require mathematics. The notation may become arcane; that is not evidence that physicists are trying to hide the subject behind priestly language.

Perhaps a simpler formulation of nature remains to be found. One can, as Feynman joked, square every equation of physics, add the squares, and set the sum equal to zero. That produces one equation, because every square must then vanish, but it produces no new understanding. Simplicity means finding the organizing idea, not merely compressing the typography. When an audience member objects to the suggestion that such an idea must come from someone young, the point is readily conceded: age is irrelevant to insight.

Overview. We shall first restore the index notation needed to recognize Lorentz scalars. We then formulate four principles for a relativistic theory, repeat the charged-particle derivation in full, and finally discover gauge invariance twice: as an endpoint term in the action and as the invariance of the electromagnetic field tensor.

1.1 The Index Language Revisited

1.1.1 Why some indices are upstairs

Let

$$x^\mu = (t, x, y, z), \quad \mu = 0, 1, 2, 3, \quad (1.1)$$

with metric

$$\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1). \quad (1.2)$$

Coordinates and coordinate displacements are assigned upper indices:

$$dx^\mu = (dt, dx, dy, dz). \quad (1.3)$$

Derivatives with respect to those coordinates carry lower indices:

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu}. \quad (1.4)$$

Which family is called upper and which lower is a convention, much as the names left and right are conventions. Once chosen, however, the convention must be followed consistently because the two families transform differently.

Question from the audience. When should an index be written as a superscript, and when should it be a subscript?

Response. Coordinates, displacements, and objects that transform like them are assigned upper indices. Derivatives and objects that transform like derivatives are assigned lower indices. The choice of which family is called upper is conventional, but the distinction between the two transformation laws is not. ^a

^aLecture timestamp: 00:05:03.

For a scalar function $s(x)$, the total differential is

$$ds = \frac{\partial s}{\partial x^\mu} dx^\mu = \partial_\mu s dx^\mu. \quad (1.5)$$

Written out, this is

$$ds = \frac{\partial s}{\partial t} dt + \frac{\partial s}{\partial x} dx + \frac{\partial s}{\partial y} dy + \frac{\partial s}{\partial z} dz. \quad (1.6)$$

The compact expression says exactly the same thing. It is not a new law; it is a device for seeing the scalar structure without carrying four terms through every line.

Einstein's convention is now immediate: an index repeated once upstairs and once downstairs is summed. The contraction in (1.5) is therefore a scalar. There is a deeper geometric interpretation of vectors and covectors, but that discussion is postponed until general relativity.

1.1.2 Raising an index

Every four-vector has upper- and lower-index forms related by the metric:

$$A_\mu = \eta_{\mu\nu} A^\nu. \quad (1.7)$$

With (1.2),

$$A_0 = -A^0, \quad A_m = A^m, \quad m = 1, 2, 3. \quad (1.8)$$

Thus raising or lowering an index changes the sign of the time component and does nothing to the spatial components. The same rule applies to the derivative:

$$\partial_\mu = \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right), \quad \partial^\mu = \left(-\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right). \quad (1.9)$$

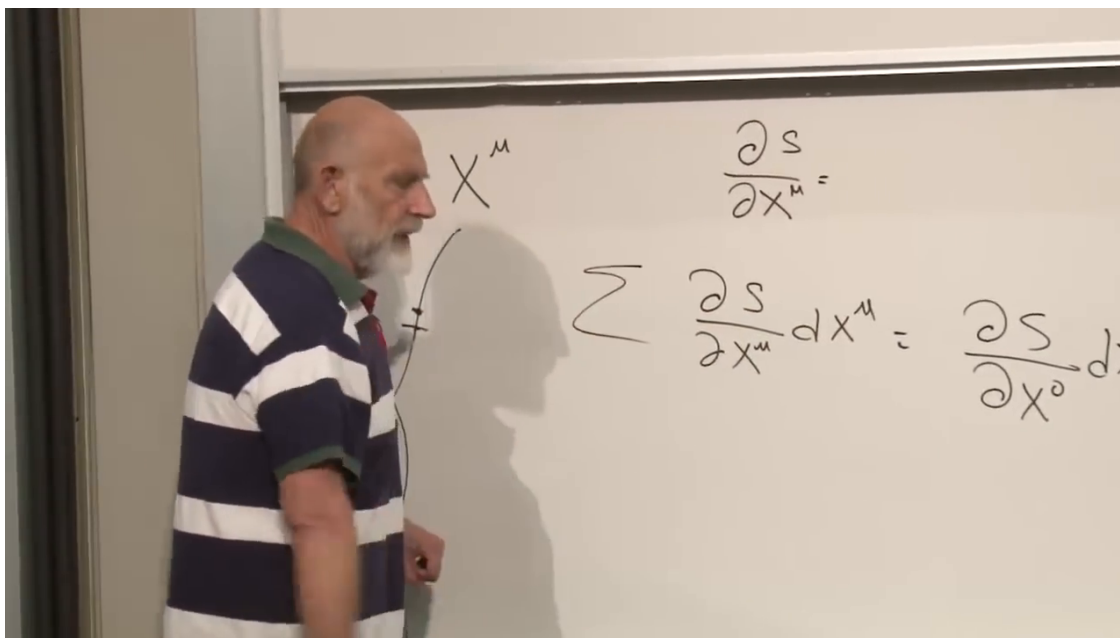


Figure 1.1: The scalar differential assembled from a lower-index gradient and an upper-index displacement.

Lecture frame: 00:08:20.

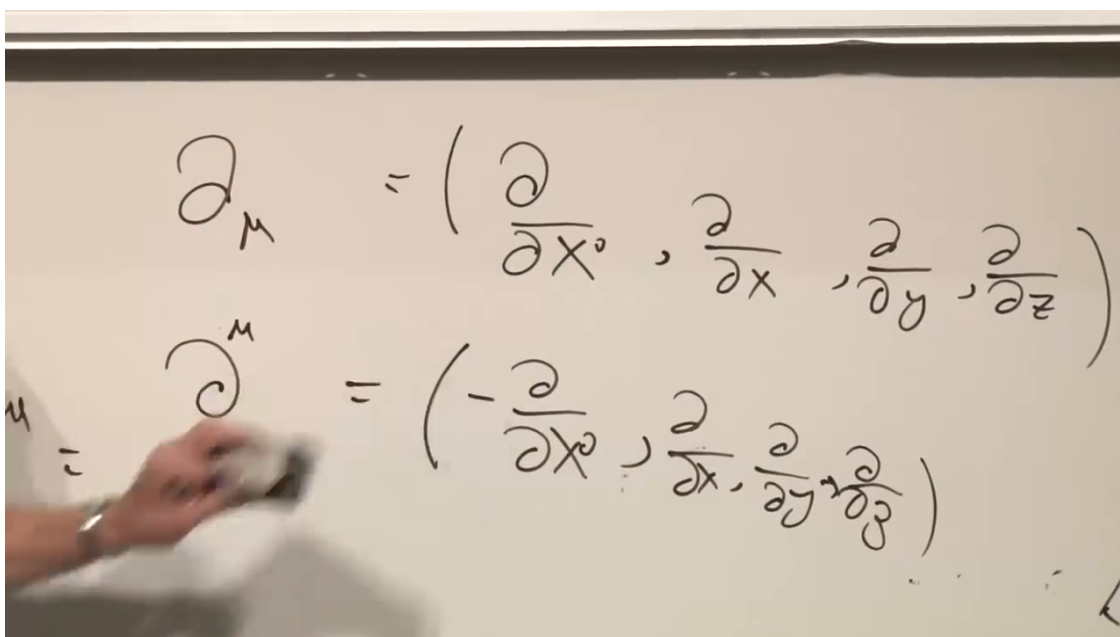


Figure 1.2: The lower- and upper-index derivative operators; only the time component changes sign.

Lecture frame: 00:12:30.

Question from the audience. Does a partial derivative always carry a lower index, and can it also be raised?

Response. The operator $\partial/\partial x^\mu$ is naturally ∂_μ , a covector. Applying the metric gives $\partial^\mu = \eta^{\mu\nu}\partial_\nu$; in flat spacetime this merely reverses the time component. ^a

^aLecture timestamp: 00:11:00.

The invariant interval is the familiar example:

$$x_\mu x^\mu = x^2 + y^2 + z^2 - t^2. \quad (1.10)$$

A contraction of one upper and one lower index produces a Lorentz scalar. This simple recognition rule is the reason for carrying the notation so carefully.

1.2 Four Principles for Constructing a Theory

Suppose an experiment reveals a new phenomenon and we want to formulate a theory of it. Four principles organize the search. They may eventually need refinement, but every successful fundamental theory presently fits their pattern.

1.2.1 The action principle

The first demand is an action. For particle mechanics,

$$I[x] = \int dt L(x, \dot{x}), \quad (1.11)$$

and the physical path is stationary under variations that hold its endpoints fixed:

$$\delta I = 0. \quad (1.12)$$

The resulting equations are

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}^m} \right) = \frac{\partial L}{\partial x^m}. \quad (1.13)$$

The action principle is not an aesthetic ornament. It is the framework that relates symmetries to conservation laws. Time-translation symmetry of an action yields energy conservation; spatial translations yield momentum conservation; related symmetries underlie other conserved quantities. The action alone is not the conservation law, but it is the structure through which the symmetry becomes one.

1.2.2 Locality

The second demand is locality. If one point of a violin string is disturbed, the neighboring segment responds first. Its neighbor responds next, and the disturbance propagates. The effect can eventually reach the whole string, but the direct short-time influence is local.

For a particle, locality is encoded by a Lagrangian evaluated along the trajectory. It depends on the position and velocity at the same time, not on arbitrary pairs of remote times. For a field $\phi(x)$,

$$I[\phi] = \int d^4x \mathcal{L}(\phi, \partial_\mu \phi). \quad (1.14)$$

The density at one event depends on the field and its first derivatives there. It does not contain a direct product of the field here with the field at an unrelated distant point.

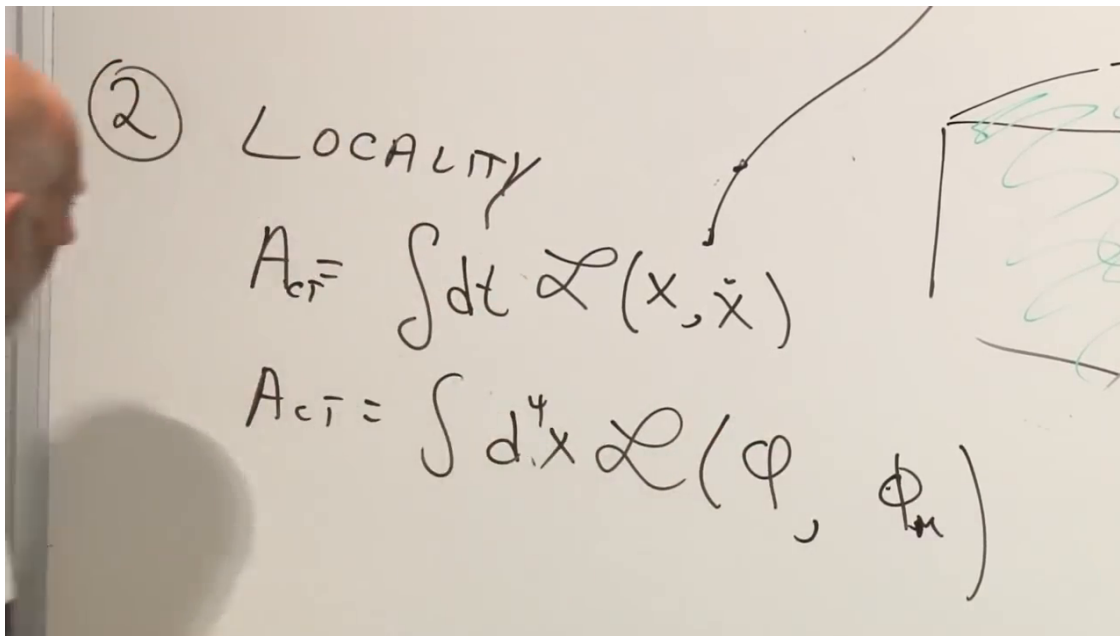


Figure 1.3: Particle and field actions displayed beneath the locality principle.

Lecture frame: 00:19:30.

This is why local equations of motion are differential equations. Derivatives compare infinitesimally neighboring values. A nonlocal alternative might relate the acceleration now to a force at a completely different time, or relate velocities at two remote times to a force at a third. Such laws are mathematically imaginable, but locality excludes them.

Question from the audience. Does quantum entanglement contradict this locality principle?

Response. Not in the dynamical sense being used here. Entanglement creates correlations, but a local operation by Alice cannot be used to alter Bob's reduced state or send him a controllable signal instantaneously. The nickel-and-dime analogy isolates the distinction: seeing one coin reveals which correlated coin the other person has, but it does not act on the distant coin. Quantum correlations are richer than that classical analogy, yet they still do not provide superluminal signaling. ^a

^aLecture timestamp: 00:20:02.

1.2.3 Lorentz invariance

The third demand is that the laws have the same form in every inertial frame. The practical construction rule is simple: the Lagrangian must be a Lorentz scalar, so the action is invariant. Upper-lower contractions such as $\partial_\mu \phi \partial^\mu \phi$ provide the building blocks.

In general relativity the statement is enlarged. The integrand is a scalar density, and invariance extends from Lorentz transformations to arbitrary coordinate transformations. That refinement does not replace the present principle; it is its curved-spacetime form.

Question from the audience. Is Lorentz invariance only a rule for special relativity, rather than a truly general principle?

Response. For a nongravitational problem in flat spacetime, Lorentz invariance is the correct statement. In curved spacetime one replaces the scalar Lagrangian by the appropriate scalar density and requires invariance under arbitrary coordinate transformations. Apart from that refinement, the same construction philosophy survives. ^a

^aLecture timestamp: 00:24:16.

Question from the audience. What are the dimensions of the particle Lagrangian and the field Lagrangian density?

Response. A particle Lagrangian has dimensions of energy. In $I = \int dt d^3x \mathcal{L}$, the spatial measure supplies a volume, so $\mathcal{L}$ has dimensions of energy per unit volume. Integrating it over space gives an ordinary Lagrangian with dimensions of energy. ^a

^aLecture timestamp: 00:25:49.

1.2.4 Gauge invariance

The fourth demand is gauge invariance. At this point it is deliberately left mysterious. An audience member offers a useful anticipation: perhaps one can add certain things to a Lagrangian without changing the physics. That is close, but the word certain will do nearly all the work.

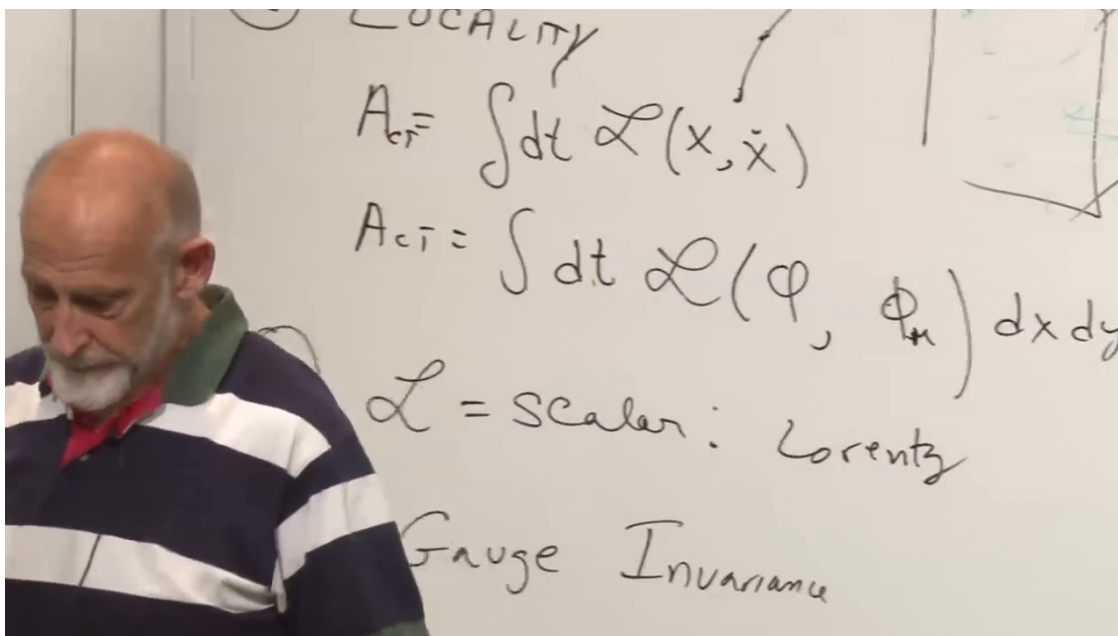


Figure 1.4: Locality, scalar construction, Lorentz invariance, and gauge invariance collected on the board.

Lecture frame: 00:27:00.

We postpone the definition because the charged-particle calculation already contains it. Repeating that calculation is not a detour. It will show why the vector potential is forced upon us and why, once introduced, it contains a redundancy.

1.3 Why the Vector Potential Is Necessary

For a slowly moving charged particle, the target equation is the familiar Lorentz-force law

$$m\mathbf{a} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (1.15)$$

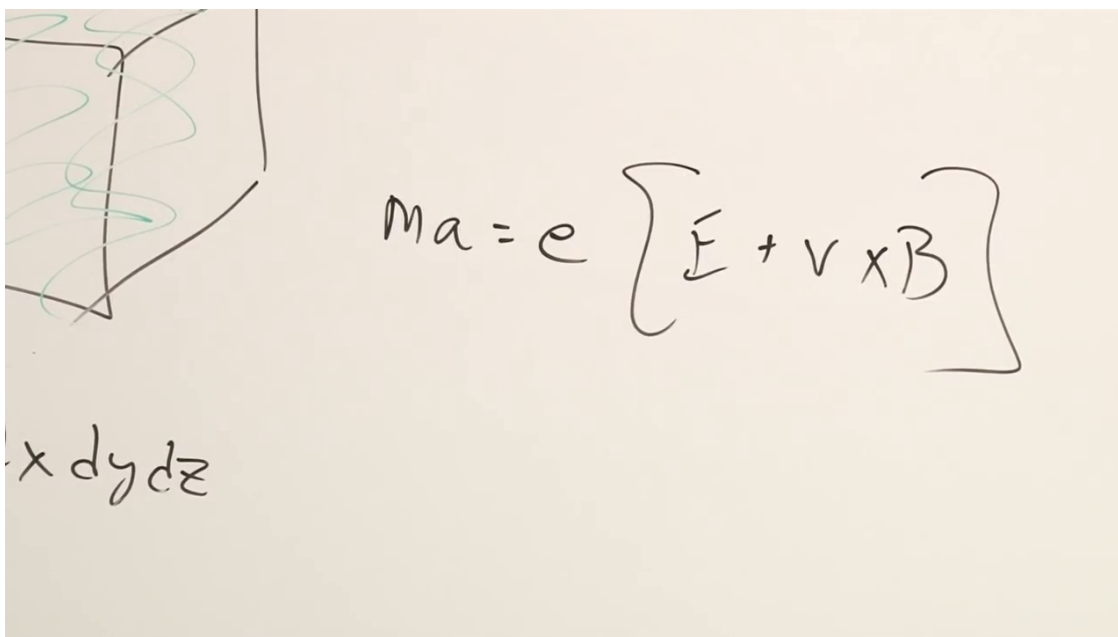


Figure 1.5: The familiar force law, written before the vector potential is reintroduced.

Lecture frame: 00:29:08.870.

No vector potential appears in (1.15). One can write the equations of motion directly with $\mathbf{E}$ and $\mathbf{B}$. The difficulty is more structural: under the assumptions of locality and Lorentz invariance used in the lecture, the interaction of a charged particle cannot be written as a worldline action directly in terms of those fields, position, and velocity alone. The potential A_μ is needed to construct the local scalar coupling.

Question from the audience. If the force law already contains only $\mathbf{E}$ and $\mathbf{B}$, why introduce A_μ ?

Response. Because an equation of motion is not yet an action principle. Within the local Lorentz-invariant framework of this lecture, the charged-particle action requires the four-potential. The fields alone can state the force, but the potential is the object that permits the force to be derived from a scalar worldline action. ^a

^aLecture timestamp: 00:29:06.

1.3.1 The worldline action

The candidate action has a free part and an interaction part:

$$I[x] = -m \int d\tau - e \int A_\mu(x) dx^\mu. \quad (1.16)$$

The first term adds the proper-time length of every infinitesimal segment of the trajectory. It is a scalar by construction. In the second term, $A_\mu(x)$ is evaluated at the event occupied by the particle and contracted with the displacement dx^μ . That increment is also a scalar and is local along the worldline.

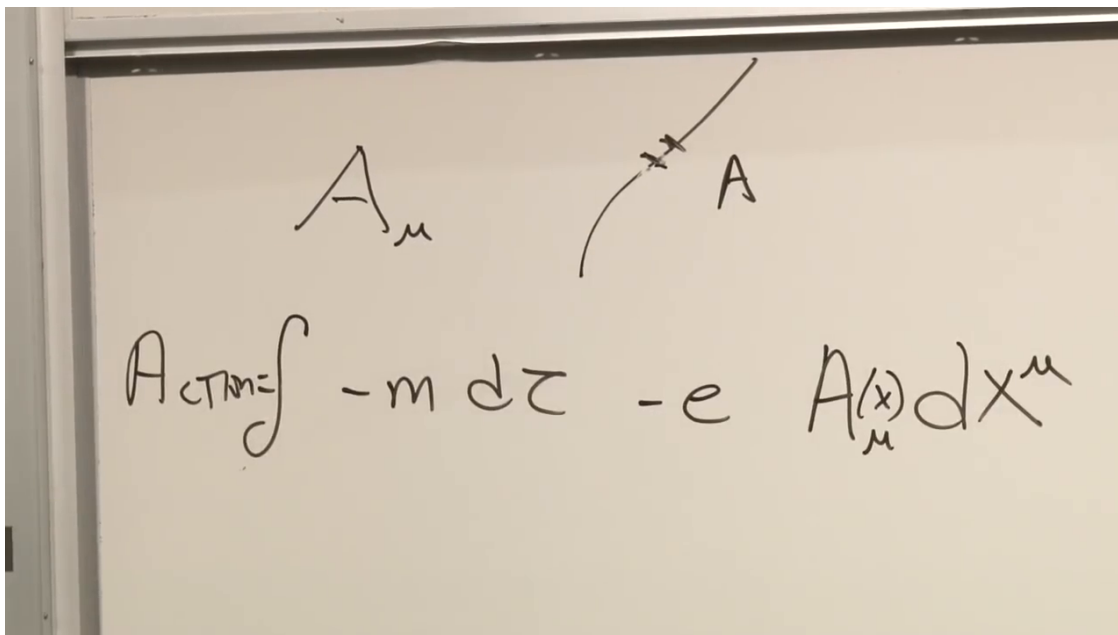


Figure 1.6: The two pieces of the charged-particle action and the local contraction $A_\mu(x)dx^\mu$.

Lecture frame: 00:31:20.

1.3.2 Rewriting the action in ordinary time

To use the familiar Euler–Lagrange equations, divide each worldline increment by dt :

$$\frac{d\tau}{dt} = \sqrt{1 - \dot{\mathbf{x}}^2}, \quad \dot{\mathbf{x}}^2 = \dot{x}^2 + \dot{y}^2 + \dot{z}^2. \quad (1.17)$$

The four components of dx^μ/dt are

$$\frac{dx^\mu}{dt} = (1, \dot{x}, \dot{y}, \dot{z}). \quad (1.18)$$

Consequently,

$$\begin{aligned} I &= \int dt \left[-m\sqrt{1 - \dot{\mathbf{x}}^2} - eA_\mu \frac{dx^\mu}{dt} \right] \\ &= \int dt \left[-m\sqrt{1 - \dot{\mathbf{x}}^2} - e(A_0 + A_m \dot{x}^m) \right]. \end{aligned} \quad (1.19)$$

The ordinary-time Lagrangian is therefore

$$L(x, \dot{x}, t) = -m\sqrt{1 - \dot{\mathbf{x}}^2} - e[A_0(x, t) + A_m(x, t)\dot{x}^m]. \quad (1.20)$$

At this stage the imaginative part is over. Once a Lagrangian has been chosen, the rest is mechanical. The mechanics can be long and tedious, but that tedium is not empty: it is what reveals the

$$\int -m \sqrt{1 - \dot{\mathbf{x}}^2} dt - e (A_0 + A_m \dot{x}^m) dt$$

$$\mathcal{L} = -m \sqrt{1 - \dot{\mathbf{x}}^2} - e (A_0(x,t) + A_m(x,t) \dot{x}^m)$$

Figure 1.7: The worldline action rewritten as an ordinary-time integral and the Lagrangian read from its integrand.

Lecture frame: 00:35:30.

electric, magnetic, and relativistic structures. The lecturer jokes that discovering the principle is the enjoyable part and deducing all its consequences is what one gives to graduate students.

Question from the audience. May the spatial index on A_m or $\dot{x}^m$ be moved upstairs or downstairs in this calculation?

Response. Yes. The spatial metric is the identity, so raising or lowering a Latin index does not change its component value. The time index is the one whose sign requires care. ^a

^aLecture timestamp: 00:34:10.

1.4 The Euler–Lagrange Calculation

1.4.1 Canonical momentum and a live correction

Choose one spatial component x^m . Differentiating the free term of (1.20) gives

$$\frac{m \dot{x}_m}{\sqrt{1 - \dot{\mathbf{x}}^2}} = m u_m, \quad u^\mu \equiv \frac{dx^\mu}{d\tau}. \quad (1.21)$$

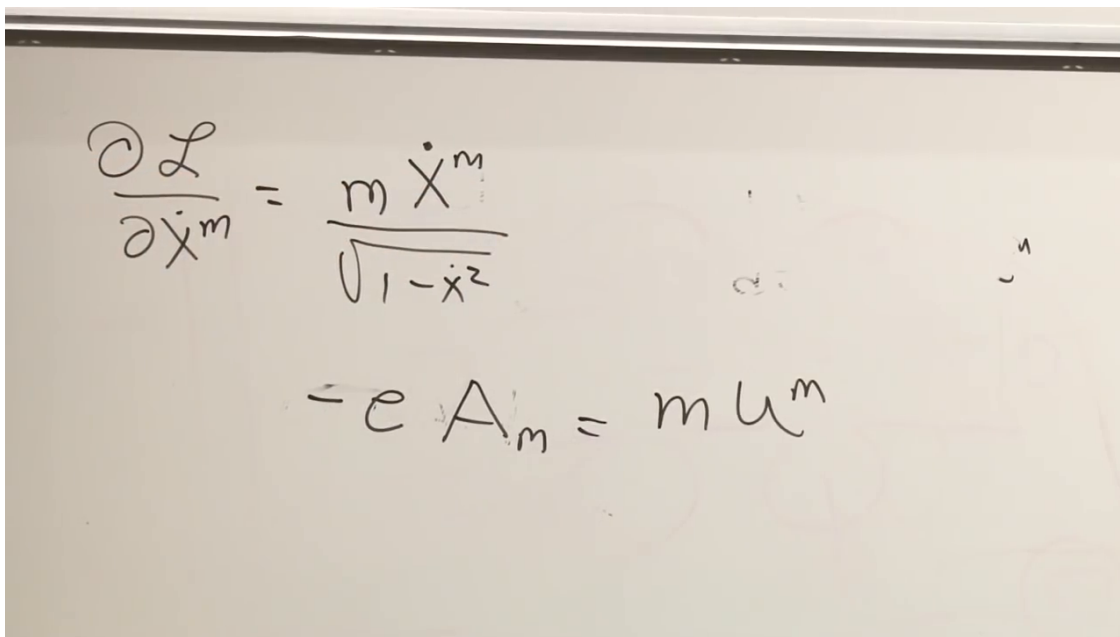
The first board attempt stops there. An audience correction catches the omission: the interaction term also depends on $\dot{x}^m$. The complete derivative is

$$\frac{\partial L}{\partial \dot{x}^m} = \frac{m \dot{x}_m}{\sqrt{1 - \dot{\mathbf{x}}^2}} - e A_m = m u_m - e A_m. \quad (1.22)$$

Question from the audience. Is the first expression for $\partial L / \partial \dot{x}^m$ missing the contribution from the electromagnetic term?

Response. Yes. Differentiating $-eA_n \dot{x}^n$ with respect to $\dot{x}^m$ contributes $-eA_m$. The corrected canonical momentum is $mu_m - eA_m$, not the mechanical momentum alone. ^a

^aLecture timestamp: 00:37:24.



$$\frac{\partial \mathcal{L}}{\partial \dot{x}^m} = \frac{m \dot{x}^m}{\sqrt{1 - \dot{x}^2}}$$

$$-e A_m = m u^m$$

Figure 1.8: The corrected canonical momentum with both the mechanical and potential terms.

Lecture frame: 00:38:30.

1.4.2 Differentiating along the moving path

The left side of the Euler–Lagrange equation is

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}^m} \right) = m \frac{du_m}{dt} - e \frac{dA_m(x(t), t)}{dt}. \quad (1.23)$$

The potential changes for two reasons: it may depend explicitly on time, and the particle moves through its spatial variation. The chain rule gives

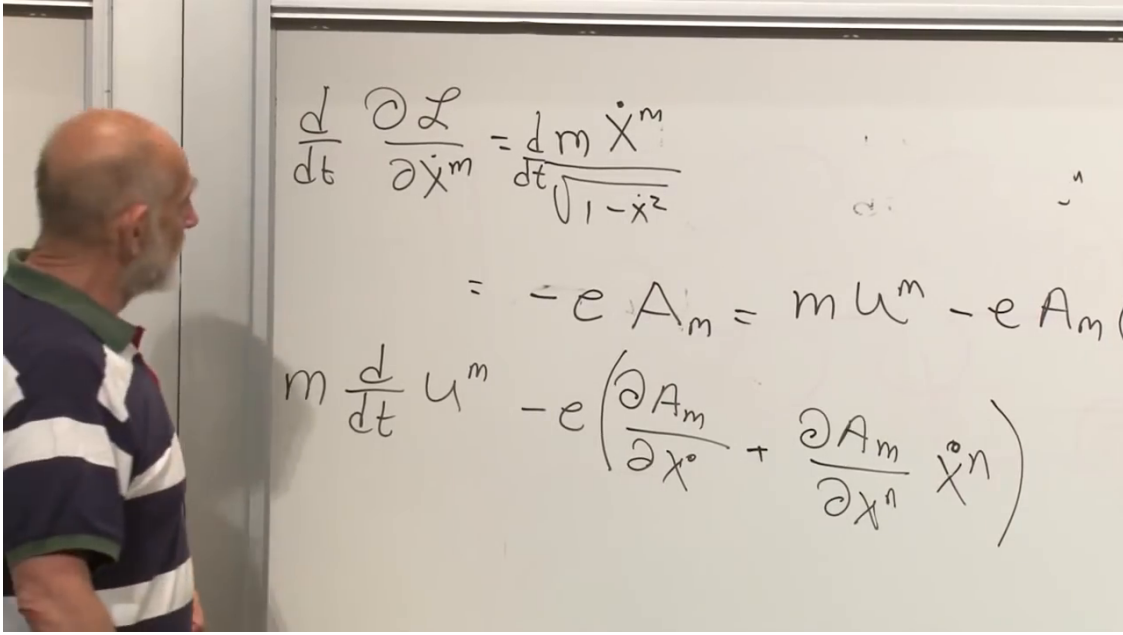
$$\frac{dA_m}{dt} = \partial_0 A_m + \dot{x}^n \partial_n A_m. \quad (1.24)$$

In longhand, the second term means

$$\dot{x} \frac{\partial A_m}{\partial x} + \dot{y} \frac{\partial A_m}{\partial y} + \dot{z} \frac{\partial A_m}{\partial z}. \quad (1.25)$$

On the right side,

$$\frac{\partial L}{\partial x^m} = -e \partial_m A_0 - e \dot{x}^n \partial_m A_n. \quad (1.26)$$


 Figure 1.9: The explicit and convective pieces of the time derivative of A_m .

Lecture frame: 00:42:10.

The free term has no explicit position dependence. In (1.26), m is the free index selecting one equation; n is a repeated dummy index and is summed. Renaming that dummy index changes nothing. This distinction is why the lecturer pauses over the letters before combining the terms.

Equating the two sides and rearranging yields

$$m \frac{du_m}{dt} = e (\partial_0 A_m - \partial_m A_0) + e \dot{x}^n (\partial_n A_m - \partial_m A_n). \quad (1.27)$$

1.4.3 Recognizing electric and magnetic fields

The first bracket contains no particle velocity. In the convention of this action, define

$$E_m \equiv \partial_0 A_m - \partial_m A_0. \quad (1.28)$$

It has precisely the operational role of the electric field. The second bracket is antisymmetric in its spatial indices:

$$C_{mn} \equiv \partial_n A_m - \partial_m A_n. \quad (1.29)$$

Its three independent components are the curl of the spatial potential, with orientation chosen in the lecture convention so that

$$\dot{x}^n C_{mn} = (\mathbf{v} \times \mathbf{B})_m. \quad (1.30)$$

The detailed component check is left as an exercise on the board. The structural identification is already visible: an antisymmetric spatial derivative of a vector has three components, exactly the number carried by $\mathbf{B}$, and it multiplies the velocity.

Equation (1.27) has therefore recovered the Lorentz force from the vector-potential action. It has also done something more important: it has exposed the antisymmetrized derivatives that will become the field tensor.

$$m a^m = e \left(\frac{\partial A_m}{\partial x^0} - \frac{\partial A_0}{\partial x^m} \right) + e \left(\frac{\partial A_m}{\partial x^\eta} - \frac{\partial A_\eta}{\partial x^m} \right) \dot{x}^\eta$$

Figure 1.10: The force separated into a velocity-independent bracket and a velocity-linear antisymmetric bracket.

Lecture frame: 00:49:30.

$$m a^m = e \left(\frac{\partial A_m}{\partial x^0} - \frac{\partial A_0}{\partial x^m} \right) + e \left(\frac{\partial A_m}{\partial x^\eta} - \frac{\partial A_\eta}{\partial x^m} \right) \dot{x}^\eta$$

$e(E_m)$ $e(B)_m$

Figure 1.11: The two derivative brackets identified as the electric and magnetic parts of the Lorentz force.

Lecture frame: 00:52:00.

1.5 From Ordinary Time to a Four-Vector Equation

1.5.1 The proper-time conversion

The left side of (1.27) is not the Newtonian acceleration $d\mathbf{v}/dt$. It is the ordinary-time derivative of a four-velocity component:

$$\frac{du^m}{dt} = \frac{du^m}{d\tau} \frac{d\tau}{dt}. \quad (1.31)$$

Multiply the equation by $dt/d\tau$. Then

$$\dot{x}^n \frac{dt}{d\tau} = \frac{dx^n}{dt} \frac{dt}{d\tau} = u^n, \quad \frac{dt}{d\tau} = u^0. \quad (1.32)$$

The spatial equation becomes

$$m \frac{du^m}{d\tau} = e (\partial_0 A_m - \partial_m A_0) u^0 + e (\partial_n A_m - \partial_m A_n) u^n. \quad (1.33)$$

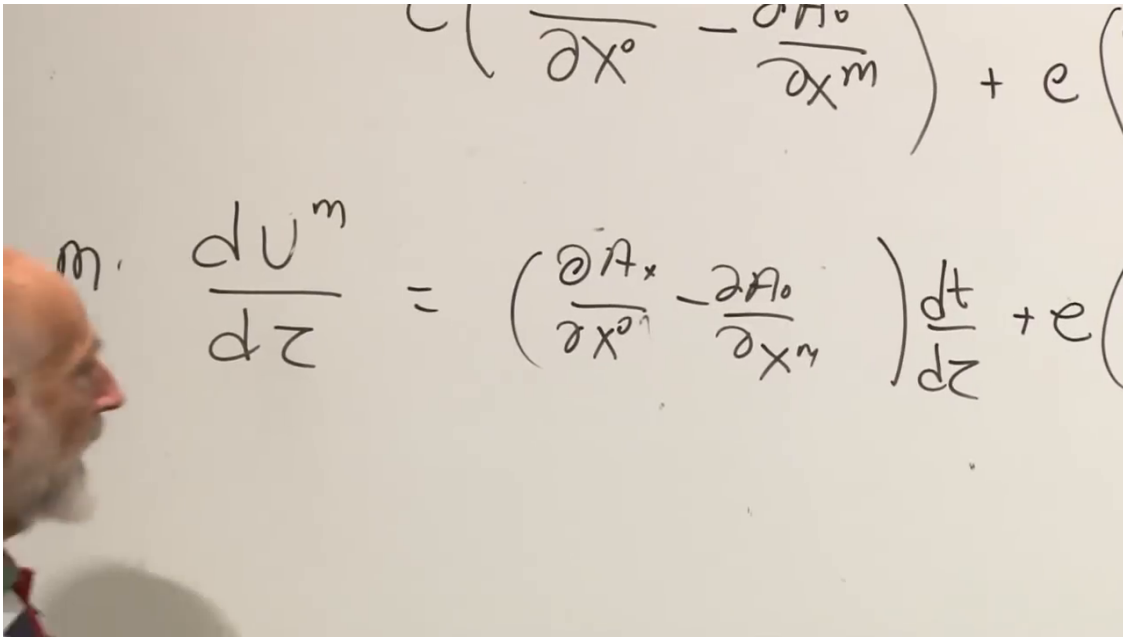


Figure 1.12: The factors $dt/d\tau$ convert ordinary velocities into components of the four-velocity.

Lecture frame: 00:56:30.

The left side is proper acceleration:

$$\frac{du^\mu}{d\tau} = \frac{d^2 x^\mu}{d\tau^2}. \quad (1.34)$$

It is a four-vector. The right side must therefore be assembled from a rank-two tensor acting on u^ν .

1.5.2 The field tensor and the force law

Choose the convention

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (1.35)$$

It is antisymmetric:

$$F_{\mu\nu} = -F_{\nu\mu}. \quad (1.36)$$

Using the interaction sign in (1.16), the compact equation is

$$m \frac{du^\mu}{d\tau} = -e F^\mu{}_\nu u^\nu. \quad (1.37)$$

Equivalently, before naming $F_{\mu\nu}$,

$$m \frac{du^\mu}{d\tau} = e (\partial_\nu A^\mu - \partial^\mu A_\nu) u^\nu. \quad (1.38)$$

Different metric, charge, and potential conventions redistribute the overall signs. The invariant content is the antisymmetrized derivative of the potential and its contraction with the four-velocity.

The image shows two handwritten equations on a piece of paper. The top equation is:

$$m \cdot \frac{du^\mu}{d\tau} = e \left(\frac{\partial A_\mu}{\partial x^\nu} - \frac{\partial A_\nu}{\partial x^\mu} \right) u^\nu + e$$

The bottom equation is:

$$m \frac{d^2 x^\mu}{d\tau^2} = e F_{\mu\nu} u^\nu$$

Figure 1.13: The proper-time equation packaged as a rank-two field tensor acting on the four-velocity.

Lecture frame: 00:58:25.

Question from the audience. The derivation produced three spatial equations. Why is a fourth equation allowed to be added?

Response. The scalar action guarantees a Lorentz-covariant result. Once the difference between the two sides is recognized as a four-vector, Lorentz transformations mix its spatial and temporal components. A four-vector equation cannot consistently have only three components; the time component accompanies the spatial ones. ^a

^aLecture timestamp: 00:59:40.

Question from the audience. Is a fixed list of numbers such as $(1, 2, 3, 4)$ itself a Lorentz-invariant four-vector?

Response. No. Such numbers can be the components of a vector in one chosen frame, but the components transform. Even in two dimensions, the vector with components $(1, 2)$ can be described in a rotated frame as $(0, \sqrt{5})$. The geometric vector persists; its numerical component list does not. ^a

^aLecture timestamp: 01:00:28.

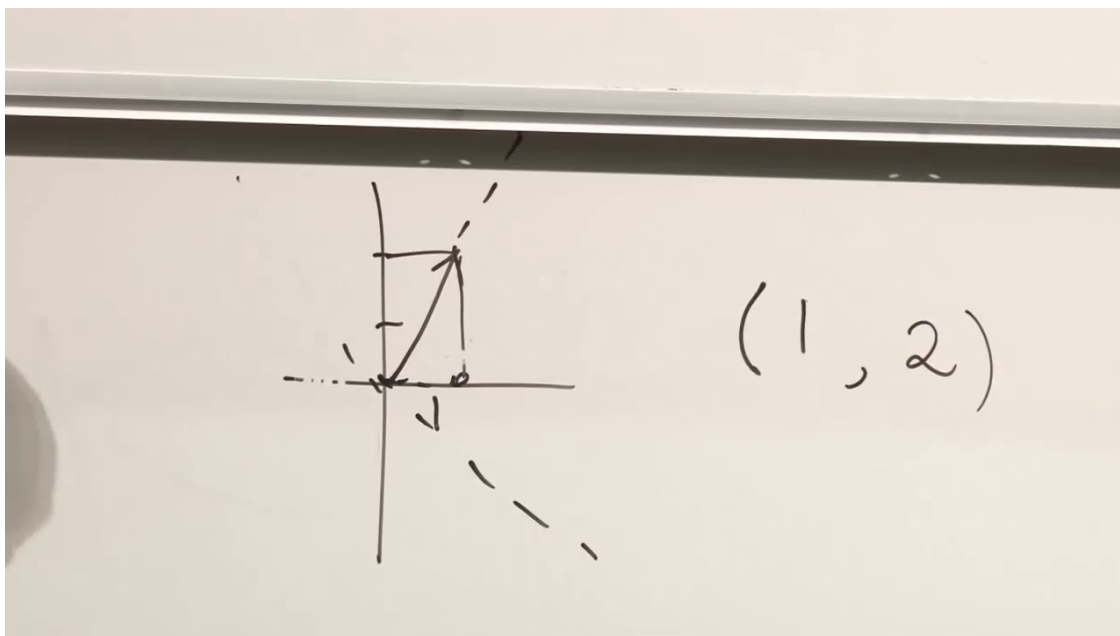


Figure 1.14: A two-component vector re-expressed in axes aligned with the vector itself.

Lecture frame: 01:02:00.

Question from the audience. If the goal was to replace ordinary time by proper time, why does $\partial_0 A_m = \partial A_m / \partial t$ still appear?

Response. The goal was not to erase every letter t . It was to write the dynamics in manifest four-vector notation. The operator ∂_0 is the time component of the covariant four-gradient, just as ∂_m are its spatial components. It belongs inside the four-dimensional tensor. ^a

^aLecture timestamp: 01:05:10.

The lecture uses this moment to restate locality. A local equation relates the acceleration at an event to the field and velocity at that event. It does not link the acceleration now to a force at a remote time.

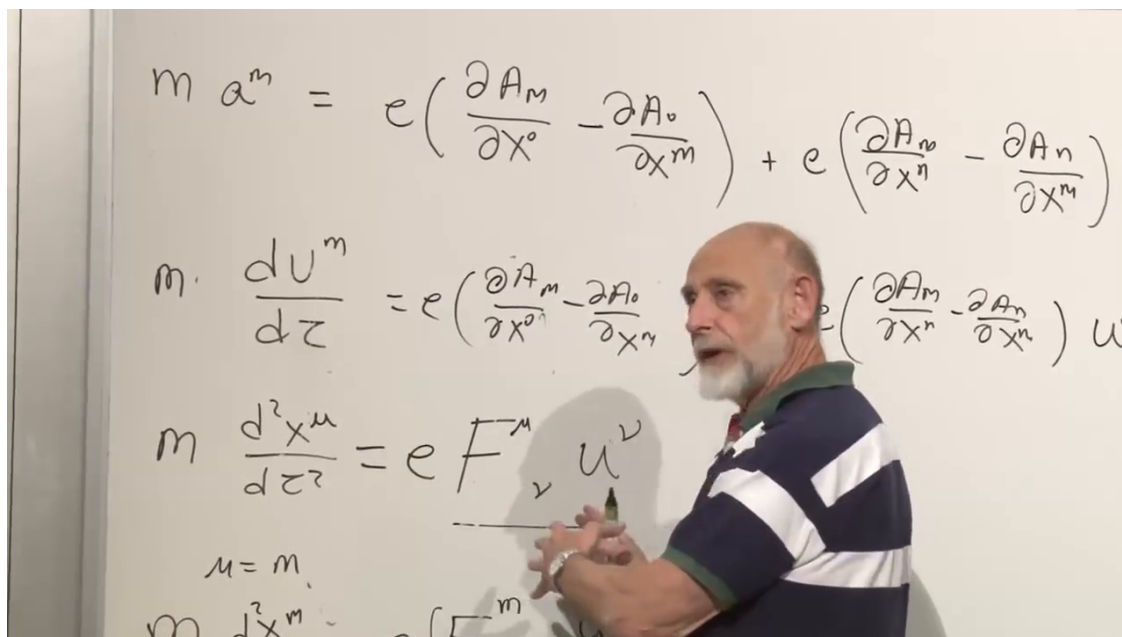


Figure 1.15: The covariant force equation used to restate locality as differential dynamics.

Lecture frame: 01:05:50.

1.5.3 Displaying the electromagnetic components

For the all-upper tensor, the convention used on the board gives

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}. \quad (1.39)$$

The row index is μ , the column index is ν . The mixed time–space entries contain the electric field; the space–space block contains the magnetic field. Antisymmetry sets the diagonal to zero and reduces sixteen displayed entries to six independent components.

Question from the audience. Is there a standard rule deciding which tensor index labels rows and which labels columns?

Response. It is a display convention. Here the first index μ runs vertically and the second index ν runs horizontally. Once that convention is announced, every component can be read unambiguously. Raising a time index introduces a sign, but it does not alter the row–column convention. ^a

^aLecture timestamp: 01:09:01.

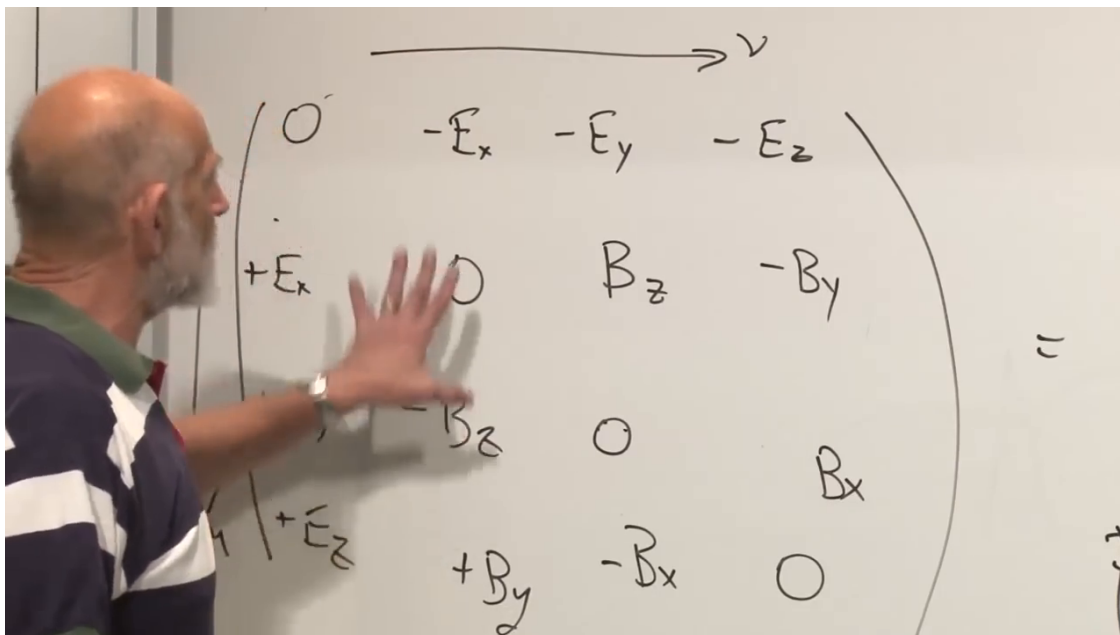


Figure 1.16: The completed electromagnetic tensor, with electric entries in the time–space slots and magnetic entries in the spatial block.

Lecture frame: 01:12:10.

Question from the audience. Which index placements preserve antisymmetry?

Response. The tensors $F_{\mu\nu}$ and $F^{\mu\nu}$ are antisymmetric. The mixed array $F^\mu{}_\nu$ is not antisymmetric under ordinary matrix transposition: raising one time index changes the electric slots, making the time–space pair symmetric, while the purely spatial magnetic block remains antisymmetric. The lecturer reaches this conclusion after briefly testing and rejecting the opposite possibility on the board. ^a

^aLecture timestamp: 01:13:18.

1.6 What Counts as an Invariance?

Only now do we return to the fourth principle. An invariance is a change that leaves the equations of motion unchanged. The action itself may remain exactly unchanged, or it may change by a boundary term that has no effect on the variational equations.

1.6.1 Constant shifts of a massless scalar

Consider the massless scalar equation in one spatial dimension,

$$\frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2} = 0, \quad (1.40)$$

obtained from

$$\mathcal{L}_0 = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi. \quad (1.41)$$

If ϕ is a solution, then so is

$$\phi \longrightarrow \phi + c, \quad (1.42)$$

where c is constant. Every derivative of c vanishes, so neither the equation nor the derivative-only Lagrangian changes.

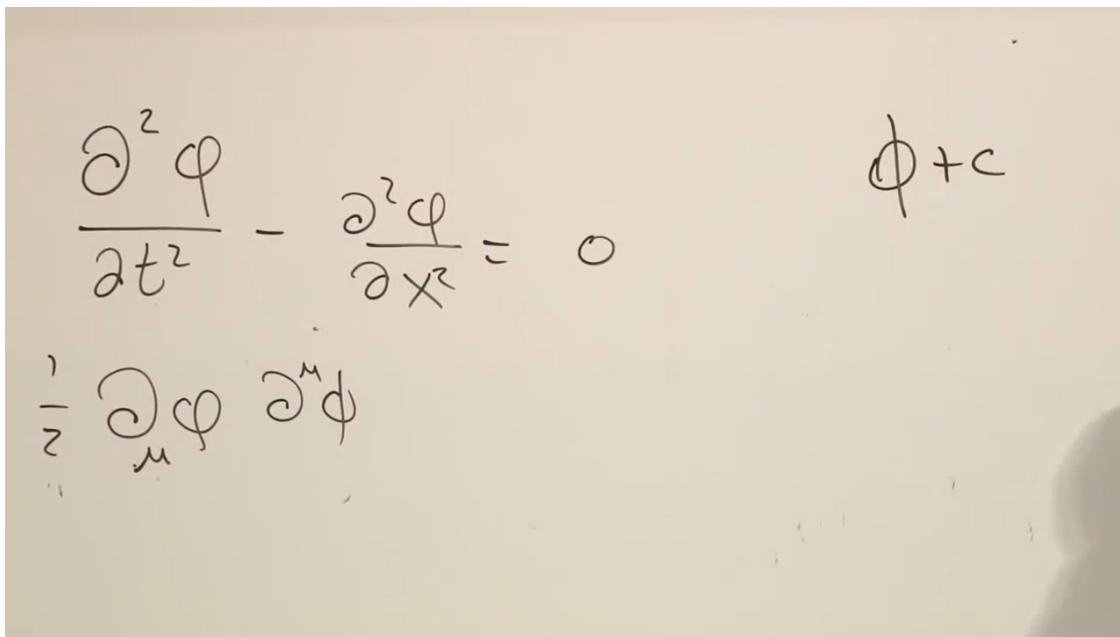


Figure 1.17: The massless wave equation, its derivative Lagrangian, and the constant shift $\phi \mapsto \phi + c$.

Lecture frame: 01:17:10.

Now add a mass term:

$$\frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2} + \mu^2 \phi = 0, \quad (1.43)$$

$$\mathcal{L}_\mu = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{\mu^2}{2} \phi^2. \quad (1.44)$$

The derivative part remains unchanged under $\phi \mapsto \phi + c$, but ϕ^2 does not. Both theories are legitimate; one has the shift invariance and the other does not.

1.6.2 A student anticipates the boundary term

An audience member proposes a sharper test:

$$\phi(x, t) \longrightarrow \phi(x, t) + x. \quad (1.45)$$

The second derivatives of x vanish, so the massless wave equation is still satisfied. The action requires a closer look. Since x and t are independent variables,

$$\frac{\partial x}{\partial t} = 0, \quad \frac{\partial x}{\partial x} = 1. \quad (1.46)$$

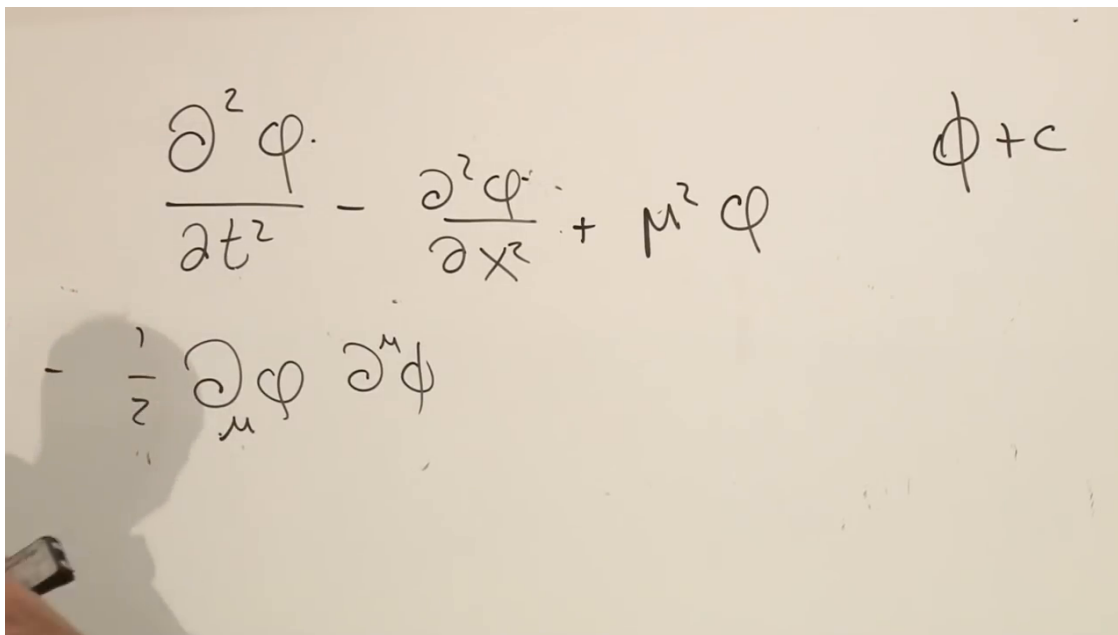


Figure 1.18: The mass term that breaks constant-shift invariance.

Lecture frame: 01:18:50.

The spatial part changes according to

$$\begin{aligned} -\frac{1}{2} [\partial_x(\phi + x)]^2 &= -\frac{1}{2} (\partial_x \phi + 1)^2 \\ &= -\frac{1}{2} (\partial_x \phi)^2 - \partial_x \phi - \frac{1}{2}. \end{aligned} \quad (1.47)$$

The new pieces are a constant and a total derivative. The constant does not change the Euler–Lagrange equation. Integrating $\partial_x \phi$ produces only spatial boundary values, so under the usual fixed-boundary conditions it also leaves the equation of motion unchanged.

Question from the audience. Does $\phi \mapsto \phi + x$ preserve the massless field equation even though it changes the Lagrangian density?

Response. Yes. The second derivatives of x vanish, so the equation is unchanged. The Lagrangian changes only by a constant and a total spatial derivative. These alter the numerical action by a fixed bulk constant and boundary values, but not the Euler–Lagrange equation under fixed boundary conditions. ^a

^aLecture timestamp: 01:20:00.

Question from the audience. Why is $\partial x / \partial t$ zero?

Response. Because x and t are independent coordinates of the field description. A partial derivative with respect to t holds x fixed. ^a

^aLecture timestamp: 01:22:22.

The student has arrived at the device needed for electromagnetism: a total derivative can change the written action without changing its stationary paths.

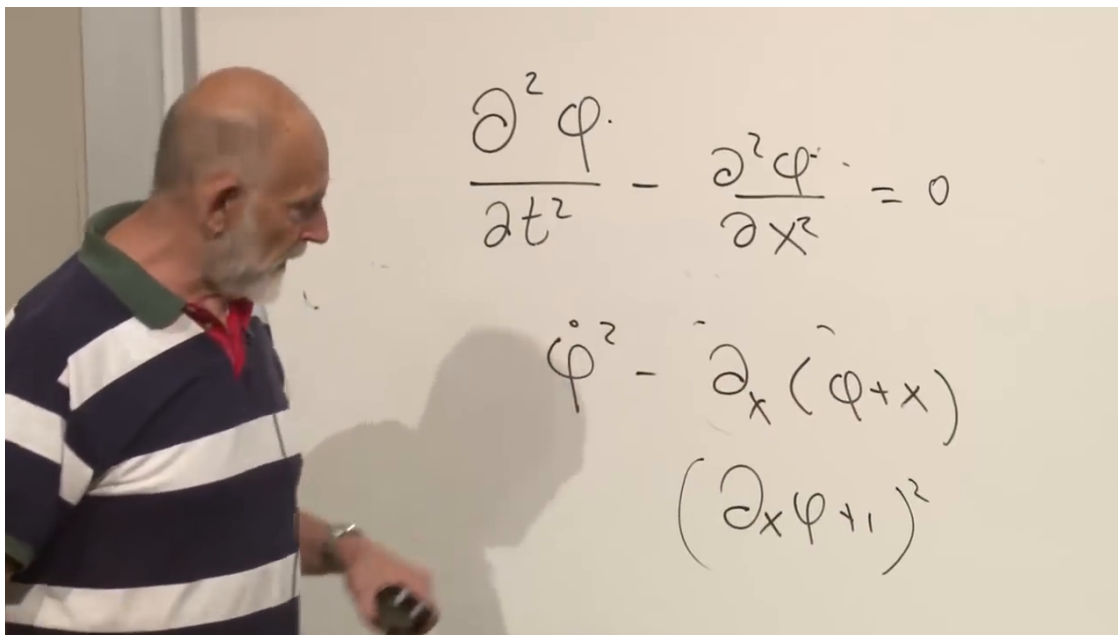


Figure 1.19: The proposed shift $\phi \mapsto \phi + x$ and the derivative square that reveals a constant plus a total derivative.

Lecture frame: 01:22:40.

1.7 Gauge Invariance of the Charged Particle

1.7.1 Adding a four-gradient

Return to the interaction

$$I_{\text{int}} = -e \int A_\mu dx^\mu. \quad (1.48)$$

Let $\Lambda(x)$ be any sufficiently smooth scalar and change the potential by

$$A_\mu \longrightarrow A'_\mu = A_\mu + \partial_\mu \Lambda. \quad (1.49)$$

The gradient $\partial_\mu \Lambda$ has the same covariant index structure as A_μ , so the sum is well defined.

The change in the action is

$$\begin{aligned} \Delta I_{\text{int}} &= -e \int \partial_\mu \Lambda dx^\mu \\ &= -e \int d\Lambda \\ &= -e [\Lambda(x_2) - \Lambda(x_1)]. \end{aligned} \quad (1.50)$$

Along each infinitesimal worldline segment, $\partial_\mu \Lambda dx^\mu$ is precisely the increment $d\Lambda$. Adding the increments telescopes to the difference between the final and initial endpoint values.

The variational principle compares nearby trajectories with x_1 and x_2 held fixed. The extra term in (1.50) is therefore the same for every allowed variation:

$$\delta \Delta I_{\text{int}} = 0. \quad (1.51)$$

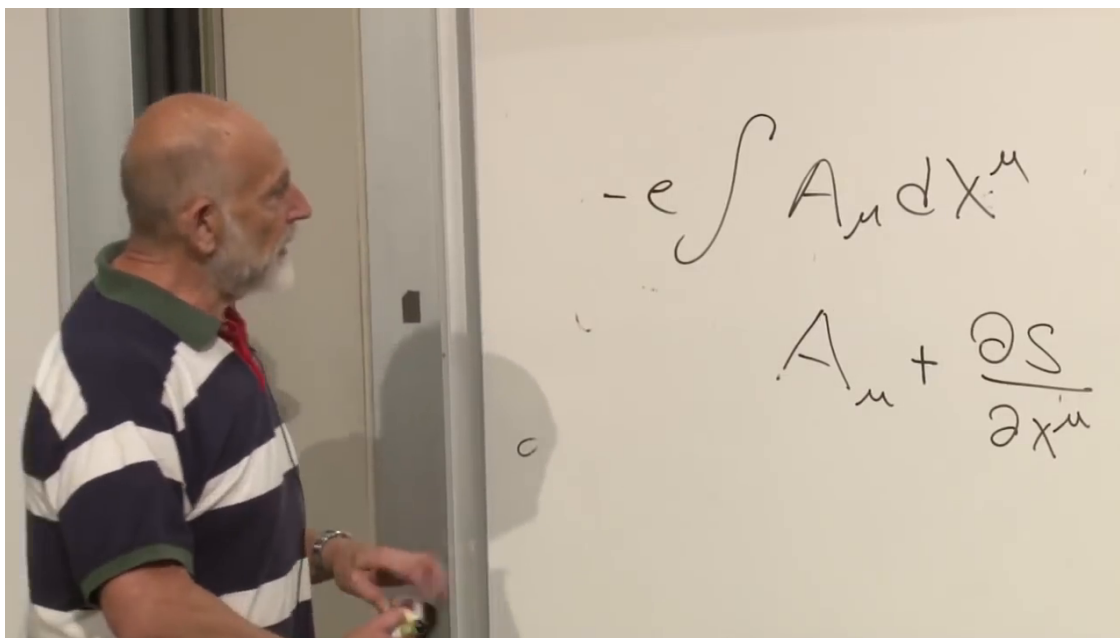


Figure 1.20: The worldline coupling and the transformation $A_\mu \mapsto A_\mu + \partial_\mu \Lambda$.

Lecture frame: 01:24:50.

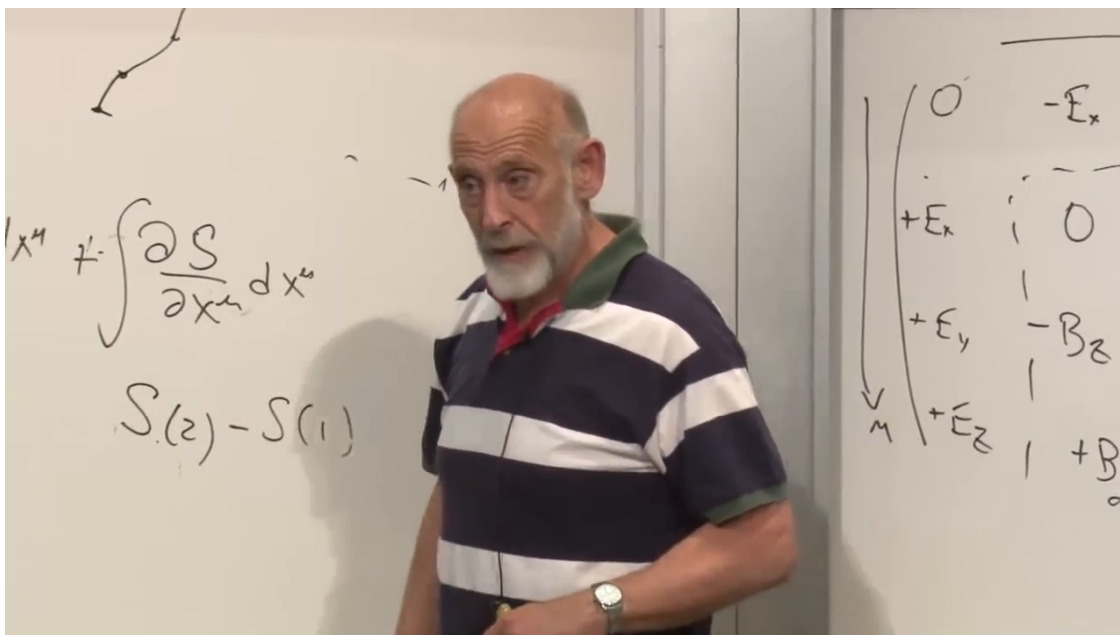


Figure 1.21: The gradient contribution integrated along the path to the endpoint difference $\Lambda(x_2) - \Lambda(x_1)$.

Lecture frame: 01:28:10.

The stationary trajectory is unchanged. This is gauge invariance as an endpoint statement.

Question from the audience. May Λ be any scalar function?

Response. Yes, subject to the smoothness and boundary assumptions needed for the derivatives and action. The transformation is local because the parameter is an arbitrary function of spacetime, not merely one global constant. ^a

^aLecture timestamp: 01:28:40.

Adding a gradient to A_μ is called a gauge transformation. The word gauge is a historical inheritance; the lecture warns, humorously but firmly, against pronouncing it as though it were a French word. The name is less important than the operation.

1.7.2 The direct proof with the field tensor

The equations of motion depend on $F_{\mu\nu}$, not on the potential by itself. Under (1.49),

$$\begin{aligned} F'_{\mu\nu} &= \partial_\mu (A_\nu + \partial_\nu \Lambda) - \partial_\nu (A_\mu + \partial_\mu \Lambda) \\ &= F_{\mu\nu} + \partial_\mu \partial_\nu \Lambda - \partial_\nu \partial_\mu \Lambda \\ &= F_{\mu\nu}. \end{aligned} \tag{1.52}$$

For a smooth scalar in ordinary flat spacetime, partial derivatives commute. The two additional terms cancel. Thus the field tensor, the electric and magnetic fields it contains, and the force law are all gauge invariant.

The image shows a handwritten derivation on a piece of paper. It starts with the definition of the field tensor $F_{\mu\nu} = \frac{\partial A_\mu}{\partial x^\nu} - \frac{\partial A_\nu}{\partial x^\mu}$. Then, it adds the terms from a gauge transformation: $+ \frac{\partial}{\partial x^\nu} \frac{\partial \Lambda}{\partial x^\mu} - \frac{\partial}{\partial x^\mu} \frac{\partial \Lambda}{\partial x^\nu}$. The final part of the derivation shows that these two terms cancel each other out because partial derivatives commute, leaving the original field tensor unchanged.

Figure 1.22: The two second derivatives generated by a gauge transformation cancel because partial derivatives commute.

Lecture frame: 01:33:50.

The two proofs close the same circle from opposite directions. The action changes only at fixed endpoints, while the equations depend only on an unchanged field tensor.

1.8 What the Redundancy Means

1.8.1 Why keep an object that is not unique?

The natural objection is immediate. If A_μ is not unique, why did we introduce it? Because it is indispensable to the local Lorentz-invariant action even though its component values contain redundant information. Many potentials represent the same electromagnetic field:

$$A_\mu \sim A_\mu + \partial_\mu \Lambda. \quad (1.53)$$

The equivalence class is physical; a particular representative is a choice of description.

Question from the audience. Is there practical value in changing the gauge?

Response. Yes. A suitable Λ can simplify a component of the potential or make one feature of a problem transparent. Locally, and with suitable boundary conditions, one can often choose a gauge in which one selected component, including A_0 , vanishes. Another gauge may display a different feature more clearly. ^a

^aLecture timestamp: 01:35:43.

Question from the audience. Is gauge invariance analogous to describing the same experiment from two Lorentz frames?

Response. No. Lorentz transformations compare physically meaningful reference frames carried by different observers. Gauge transformations are redundancies within one description: different potentials encode the same field. Mathematically both are invariances, but their interpretations differ. ^a

^aLecture timestamp: 01:37:07.

There is also a structural difference. A rotation uses a fixed global angle; one does not ordinarily rotate the axes by unrelated angles at each point. A gauge transformation is parameterized by the entire function $\Lambda(x)$. This position-dependent freedom is what makes it a local gauge invariance.

Question from the audience. Is adding a gradient the general definition of gauge invariance?

Response. For the Abelian electromagnetic theory, that is the transformation. Yang–Mills theories have a more elaborate non-Abelian version in which the potential carries additional internal structure, but the same idea survives: different local descriptions represent the same physics. ^a

^aLecture timestamp: 01:39:27.

Question from the audience. Is a gauge transformation merely a mathematical device for highlighting different parts of a problem?

Response. It is a useful calculational and interpretive device, but its role is constrained: it may simplify the equations or reveal a property, yet it cannot change any final gauge-invariant prediction. ^a

^aLecture timestamp: 01:40:22.

1.8.2 Other fields and other potentials

Gauge potentials in other theories are also commonly denoted by A and carry a spacetime index μ . What distinguishes them is additional internal structure associated with other symmetries. The lecture postpones that structure rather than offering a premature formula.

Question from the audience. What do the vector potentials of other field theories look like?

Response. They still behave as spacetime vectors and are commonly written A_μ , but they carry additional labels and algebraic structure associated with internal symmetries. That is the entrance to Yang–Mills theory and is intentionally deferred. ^a

^aLecture timestamp: 01:41:02.

Question from the audience. Is it really impossible to write the local Lorentz-invariant charged-particle action directly with $\mathbf{E}$ and $\mathbf{B}$?

Response. Within the assumptions stated here, yes. The lecture presents this as a theorem and does not prove it. Preserving the action principle, locality, and Lorentz invariance forces the introduction of the potential, after which gauge redundancy appears automatically. ^a

^aLecture timestamp: 01:42:18.

1.8.3 What can be measured?

A local value of A_μ by itself is not gauge invariant. If a physical measurement determined that value uniquely, changing it by $\partial_\mu\Lambda$ would alter a physical outcome, contrary to the invariance just proved. Measurable predictions must be built from gauge-invariant quantities, such as $F_{\mu\nu}$ in the present classical discussion.

Question from the audience. Why does nature so persistently choose gauge theories?

Response. At the classical level the lecture offers no final reason. Quantum mechanics supplies a deeper perspective, but even there the broad answer is not exhausted. What is already evident is the unusual beauty, symmetry, and economy of theories organized by gauge redundancy. ^a

^aLecture timestamp: 01:43:36.

Question from the audience. Can the vector potential itself be measured?

Response. Not as a gauge-dependent local component value. A measurement is a physical process, and every physical prediction must be insensitive to $A_\mu \mapsto A_\mu + \partial_\mu\Lambda$. What can be measured are gauge-invariant features of the potential and field. ^a

^aLecture timestamp: 01:44:18.

Question from the audience. What happens if A_μ is changed by something that is not a gradient?

Response. In general $F_{\mu\nu}$ changes, so the change is detectable. By contrast, a pure-gauge potential $A_\mu = \partial_\mu\Lambda$ has $F_{\mu\nu} = 0$ and exerts no electromagnetic force; a free particle then moves on a straight worldline. ^a

^aLecture timestamp: 01:45:22.

Gauge invariance is therefore not a disposable trick. Electrodynamics, Yang–Mills theory, the Standard Model, and gravity all possess gauge redundancies of one kind or another. The pattern is peculiar: an auxiliary quantity is required to formulate the theory, yet part of that quantity can be changed without changing the physics. The peculiarity is also the organizing power.

1.9 The Logic of Gauge Invariance

The lecture has taken a deliberately indirect path. We first restored the upper–lower index language because scalar contractions are the grammar of Lorentz-invariant actions. We then stated four principles: action, locality, Lorentz invariance, and gauge invariance. The fourth remained undefined while the charged-particle calculation was rebuilt.

That calculation began with the local scalar action

$$I = -m \int d\tau - e \int A_\mu dx^\mu, \quad (1.54)$$

passed through the ordinary-time Lagrangian and every Euler–Lagrange derivative, and recovered the electric and magnetic Lorentz force. Proper time then assembled the result into

$$m \frac{du^\mu}{d\tau} = -e F^\mu{}_\nu u^\nu, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (1.55)$$

The scalar-field examples prepared the final step: a total derivative can alter an action only through boundaries. Consequently,

$$A_\mu \longrightarrow A_\mu + \partial_\mu \Lambda \quad (1.56)$$

changes the worldline action only by fixed endpoint values and leaves $F_{\mu\nu}$ unchanged because mixed partial derivatives commute. The vector potential is therefore necessary but nonunique. Gauge invariance is the statement that this nonuniqueness belongs to the description, not to the physics.